# Chapter 5: Auctions and Mechanism Design

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# Outline


1. Motivation: Various Auction Formats
2. Second-Price Auction in Detail
3. Mechanism Design Concepts
4. Revenue Equivalence Theorem
5. Vickrey-Clarke-Groves Mechanisms
6. Wrap-Up and Exercises

# Common Auction Formats

- **First-Price Sealed-Bid**:
  - Each bidder submits a single bid; highest bid wins and pays that bid.
- **Second-Price (Vickrey) Sealed-Bid**:
  - Highest bidder wins but pays the *second-highest* bid.
- **English (Ascending)**:
  - Price starts low and rises until only one bidder remains; that bidder pays the final price.
- **Dutch (Descending)**:
  - Price starts high and decreases until someone accepts, paying that "stop" price.

**Remark**

*We focus on single-item, private-value auctions, ignoring multi-unit or common-value complications at first.*

# English vs. Dutch: Approximate Equivalence

- **English (Ascending)** Auction
  - In a private-value setting, the final price often aligns with the **second-highest** valuation. Because once the second-highest drops out, the last man stands near that threshold.
  - This is *similar* to a second-price sealed-bid format.
- **Dutch (Descending)** Auction
  - Bidders "stop the clock" at the price they're willing to pay; the winner's price is her own stop.
  - Strategically akin to a first-price sealed-bid: you effectively pay exactly what you "shout," so you shade your reveal below your true value.

**Hence**

English $\leftrightarrow$ second-price; Dutch $\leftrightarrow$ first-price in equilibrium outcomes, if risk-neutral private values.

# Private Value vs. Common Value

- **Private Value**:
  - Each bidder's valuation is independent of others.
  - E.g., collectible items, purely subjective value.
- **Common Value**:
  - The item has one true value for everyone, but each bidder has different signals.
  - E.g., oil lease with uncertain reserves; "winner's curse" arises if the winner overestimates the item's value.
- **Affiliated/Correlated**:
  - Partial correlation between bidders' signals; more general real-world scenario.

**Remark**

*We mostly handle private values. Common-value auctions require additional shading and adjusting for the* **winner's curse**.

# Second-Price Sealed-Bid Auction

- Each bidder submits a single sealed bid $b_i$.
- The highest bid wins, paying the *second*-highest bid.
- **Dominant Strategy Claim**: $b_i(v_i) = v_i$ if valuations are private (i.i.d., risk-neutral).

**Reasoning**:

- Payment depends on *other* bids, not your own.
- If you under-bid, you risk losing a profitable win (paying second-highest, which could be smaller than your $v_i$).
- If you over-bid, you risk winning but paying second-highest greater than $v_i$.
- So truth-telling is *weakly* better than any deviation.

# First-Price Sealed-Bid (Recall)

- Highest bidder wins, but pays her own bid.
- **Equilibrium** for private-value: each bidder *shades* the bid below $v_i$ to balance profit margin vs. probability of winning.
- Example: $n = 2$, uniform(0,1) $\implies b^*(v_i) = \frac{1}{2} v_i$.

So in first-price, no "dominant strategy"—your best bid depends on the distribution and number of bidders.

# n=2, Uniform(0,1) Example: First- vs. Second-Price

**First-Price**:

- BNE strategy: $b_i^*(v_i) = \frac{1}{2} v_i$.
- If $v_1 > v_2$, bidder 1 wins, paying $\frac{1}{2} v_1$.
- Probability $v_1 > v_2$ is $1/2$. Also $\mathbb{E}[v_1 \mid v_1 > v_2] = \frac{2}{3}$.
- So expected payment by winner is $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$.

**Second-Price**:

- Each bids truthfully: $b_i = v_i$.
- If $v_1 > v_2$, winner pays $v_2$. $\mathbb{E}[v_2 \mid v_2 < v_1] = \frac{1}{3}$ (similar integration).
- Summing symmetrical cases, total expected revenue $= \frac{1}{3}$.

## Conclusion

They yield the *same* expected revenue $\frac{1}{3}$.

# Is This A Coincidence?

- We see first-price and second-price yield the same *average* revenue for $(n = 2,\ \text{Uniform}(0, 1))$.
- Under broader conditions (risk-neutral, i.i.d., always allocate to highest $v$), we'll discover the **Revenue Equivalence Theorem** that generalizes this result.

**But first, let's step back to the mechanism design viewpoint.**

We want to see auctions as special "mechanisms" for resource allocation.

# Mechanism Design: A Broader Definition

- Not just auctions: **any** procedure to allocate resources or decide outcomes under private information, aiming for:
  - Efficiency (maximize sum of valuations),
  - Or revenue (seller's profit),
  - Or fairness, etc.
- **Examples**:
  - Setting up a contract menu for employees or customers,
  - Government deciding how to finance and provide public goods,
  - Matching markets (e.g., hospitals-residents) with preference lists.
- **Key Tools**: Revelation principle, IC, IR, design of payment rules.

# Mechanism Design: Broad Scope

- A **mechanism**:
  - Agents with private types $\theta_i$ (e.g., valuations).
  - Each "reports" (message) $\hat{\theta}_i$ (which may differ from $\theta_i$).
  - Mechanism decides an outcome $o(\hat{\theta})$ and, often, transfers (who pays how much).
- It's **not** just auctions:
  - A company setting different prices for different consumer segments is effectively creating a "screening mechanism."
  - Government designing a pollution permit system is also a mechanism.
  - So it's a very general notion.

# Revelation Principle: Key Simplifier

**Statement**:

- If a mechanism (possibly complicated) implements outcome $f(\theta)$ in equilibrium, then there is a *direct* mechanism in which each agent "reports" $\theta_i$ and telling the truth is an equilibrium that yields $f(\theta)$.

**Proof Sketch**:

- In the original mechanism, each type $t_i$ uses a strategy $\sigma_i(t_i)$.
- Build a direct mechanism: each agent $i$ reports $\hat{\theta}_i$, the mechanism simulates $\sigma_i(\hat{\theta}_i)$ to produce the same outcome.
- Telling the truth ensures you replicate the payoff of the original equilibrium. So truth-telling is an equilibrium in the direct version.

**Hence**

We can focus on **direct, truthful** mechanisms *w.l.o.g.*, analyzing *IC* and *IR* directly.

# Incentive Compatibility, Individual Rationality

**IC**: For each type $\theta_i$, truth-telling must be a best response:

$$u_i(\theta_i, \theta_i) \geq u_i(\hat{\theta}_i, \theta_i), \ \forall \hat{\theta}_i, \theta_i.$$

**IR**: For each type $\theta_i$, the utility of participating truthfully is nonnegative:

$$u_i(\theta_i, \theta_i) \geq 0, \ \forall \theta_i.$$

**One-Item Example**:

$$\text{If agent } i \text{ wins, payoff } = \theta_i - p_i(\hat{\theta}), \text{ else } 0.$$

- IC means $\theta_i - p_i(\theta_i)$ is at least as good as any deviation's outcome. - IR means $\theta_i - p_i(\theta_i) \geq 0$ for the type who actually wins (or 0 if that type never wins).

# Envelope Theorem & Payoff Derivative (Intuitive Note)

- In single-dimensional settings (like "valuation $\theta$ for a single item"), if the probability of winning for type $\theta$ is $x(\theta)$, and $p(\theta)$ is the expected payment.

$$u(\hat{\theta}, \theta) = x(\hat{\theta})\theta - p(\hat{\theta})$$

  and define $u(\theta) = u(\theta, \theta)$, the **envelope theorem** implies

$$\frac{d}{d\theta} u(\theta) = x(\theta),$$

  if the mechanism is **IC**.
- So $u(\theta) = u(0) + \int_0^\theta x(t)\, dt$.
- If $u(0) = 0$ (lowest type gets 0 utility), then $u(\theta) = \int_0^\theta x(t)\, dt$.

This is key to the proof of Revenue Equivalence coming up.

# Revenue Equivalence: Setup

- $n$ risk-neutral bidders, i.i.d. valuations $\theta \in [0, \bar{\theta}]$.
- A mechanism that **always** awards the item to the highest type.
- **Boundary condition**: A type $\theta = 0$ (lowest possible) gets $u(0) = 0$ (no surplus).
- Then we want to see if the **expected payment** from the winner is pinned down—hence "equivalence."

## Key steps

1) The payoff function $u(\theta)$ is determined by $x(\theta)$ (win probability) via an integral (envelope). 2) Summation of payoffs + total surplus is fixed by the allocation rule. 3) The seller's revenue is the leftover.

# RET: The Envelope Argument (Sketch)

- Let $x(\theta)$ = probability type $\theta$ wins (which is 1 if $\theta$ is the highest among $n$).
- Under IC, $u'(\theta) = x(\theta)$. So

$$u(\theta) = u(0) + \int_0^\theta x(t)\,dt = \int_0^\theta x(t)\,dt,$$

  if $u(0) = 0$.
- Summed across all possible $\theta$, we get total **consumer surplus** as an integral depending on $x(\theta)$.
- Meanwhile, the **total surplus** from awarding to the highest type is $\int \ldots$ a constant (given the same allocation rule).
- $\implies$ The *difference* = seller's expected revenue is fixed.

## Hence

Any mechanism that "highest type wins" + $u(0) = 0$ yields the same expected revenue. This includes first-price, second-price, English, Dutch auctions, etc.

# Final Statement of Revenue Equivalence

**Theorem (Informal)**: In a single-good, risk-neutral environment with i.i.d. private values, any mechanism that always allocates the good to the highest type and grants a type-0 bidder zero utility yields the same **expected revenue**.

**Corollary**: 1st-price, 2nd-price, English, Dutch auctions (under no reserve, no collusion, ...) have the same average revenue.

**Remark**

*Risk aversion, correlated signals, or changing the allocation rule (e.g. ignoring the highest type sometimes) can break equivalence.*

# Vickrey-Clarke-Groves (VCG) Mechanisms

- **General scenario**:
  - $n$ agents, each has valuation $v_i(o)$ over possible outcomes $o \in O$.
  - Mechanism chooses outcome $o^*$ that maximizes $\sum_i \hat{v}_i(o)$ (sum of reported valuations).
- **Payment rule**: each agent $i$ pays the "externality cost" he imposes on others:

$$p_i = \max_{o \in O} \sum_{j \neq i} \hat{v}_j(o) \; - \; \sum_{j \neq i} \hat{v}_j(o^*).$$

- **Result**:
  - *truth-telling* is a dominant strategy, ensuring that the outcome is **socially efficient**.

# Why VCG is Truthful? (Intuition)

- Each $i$'s payoff $= \hat{v}_i(o^*) - p_i$, and $p_i$ is structured so that "cheating" does not improve net utility.

$$\hat{v}_i(o^*) - p_i = \sum_i \hat{v}_i(o^*) - \max_{o \in O} \sum_{j \neq i} \hat{v}_j(o)$$

## Connection to Single-Item Auctions

- Vickrey second-price is the special case of VCG with $O =$ "which bidder gets the item, or none." - Payment formula reduces to paying the second-highest bid.

# Summary

- We saw multiple auction formats (first-price, second-price, English, Dutch).
- Under private values and risk neutrality, **revenue equivalence** emerges—different rules yield the same expected revenue if the highest valuation always wins.
- **Second-price** is special because truth-telling is a *dominant strategy*.
- **Mechanism design** generalizes these ideas:
  - **Revelation Principle** $\Rightarrow$ focus on direct, truthful mechanisms.
  - **IC, IR** $\Rightarrow$ key constraints.
  - **VCG** $\Rightarrow$ ensuring efficient outcome with truthful reports in more complex settings.

# Exercises (1)

1. **Second-Price Auction Dominant Strategy**:
   - Provide a short proof that $b_i(v_i) = v_i$ is a dominant strategy in the sealed-bid second-price auction with private values.
2. **n=2 Example Calculation**:
   - Show carefully how first-price (with $b^*(v) = \frac{1}{2}v$) yields $\frac{1}{3}$ expected revenue for uniform(0,1).
   - Show second-price also yields $\frac{1}{3}$.

# Exercises (2)

1. **Mechanism Design Basics**:
   - State the definition of IC and IR in a direct mechanism for single-item allocation.
   - Provide a small numerical example with 2 types to illustrate.
2. **Revelation Principle**:
   - Write 1-2 paragraphs constructing the direct mechanism from an indirect game, ensuring truth-telling reproduces the original equilibrium payoffs.

# Exercises (3)

1. **Revenue Equivalence Theorem (Sketch)**:
   - Suppose highest valuation always wins, $u(0) = 0$.
   - Use envelope theorem to show $u(\theta) = \int_0^\theta x(t)\,dt$.
   - Argue the leftover is the seller's revenue, hence same across all such mechanisms.
2. **VCG for 2 Items, 3 Bidders**:
   - Each bidder $i$ can have $v_i(\cdot)$ on different allocations (who gets item1, item2).
   - Write the formula for $p_i$.
   - Why is it truthful?

# Further Discussion

- **Myerson's Optimal Auction**:
  - Goes beyond awarding to highest valuation—introduces "virtual valuations," reserve prices, etc., to maximize *revenue*.
  - We don't detail it here, but it's a fundamental result in advanced mechanism design.
- **Risk Aversion, Collusion, Multi-Item** contexts:
  - RET can fail or new complexities arise (e.g., bundling synergy).
  - Real auctions (like spectrum, combinatorial) use advanced rules that might deviate from these simple results.

# References


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# Concluding Remarks

- We introduced **Second-Price** with a *dominant strategy*, contrasted with **First-Price**.
- For $n = 2$, uniform(0,1), their *expected revenues* coincide—leading to the bigger idea of **Revenue Equivalence** in private-value auctions.
- **Mechanism Design** is a powerful lens: auctions are just one example. The **Revelation Principle** simplifies analysis, and **IC/IR** conditions shape possible outcomes.
- **VCG** ensures efficiency in more general contexts, while **Myerson's Auction** addresses revenue maximization.

## Next Steps

Potential expansions: - Multi-item auctions, combinatorial synergy. - Common value settings (winner's curse). - Optimal design & advanced mechanism design frameworks.

Q & A

Thank you for your attention!